

SIMATS ENGINEERING

ASSESSMENT TOOL 1-APPLICATION- BASED QUESTIONS

COURSE NAME: Cryptography and Network Security

COURSE CODE: CSA5103

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 40%

QUESTIONS: 5

Total Marks:100

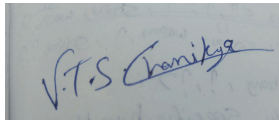
Annexure A

APPLICATION- BASED QUESTIONS

Code of Conduct:

I V.T.S.CHANIKYA(192424290) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



APPLICATION- BASED QUESTIONS

Scenario: Secure Messaging System A company deploys a keyword-based encryption technique for internal communication. An employee sends the word “HELLO” using the keyword “KEY” to ensure confidentiality.

Demonstrate how the encryption process transforms the message step-by-step.

Explain how the repeating keyword influences each character of the ciphertext.

Scenario: Suspicious Digital Asset Investigation During a digital forensic audit, investigators suspect that a company logo file may contain hidden confidential information embedded within it. The file appears normal upon casual inspection.

Analyze how investigators could detect the presence of hidden data within the image.

Recommend two techniques that could strengthen the security of such hidden communications against detection or extraction.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding g; limited or partial integration	Poor understanding g; unable to integrate concepts
Experiment al Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation on with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation

Answers – Application-Based Questions

3. Scenario: Secure Messaging System

The keyword-based encryption technique described is the **Vigenère cipher**, a classical polyalphabetic substitution cipher.

Given:

- **Plaintext:** HELLO
- **Keyword:** KEY
- **Repeated keyword:** KEYKE

Using the alphabet values **A = 0, B = 1, ..., Z = 25**:

Position	Plaintext	Value	Keyword	Value	Calculation	Ciphertext
1	H	7	K	10	$(7 + 10) \bmod 26 = 17$	R
2	E	4	E	4	$(4 + 4) \bmod 26 = 8$	I
3	L	11	Y	24	$(11 + 24) \bmod 26 = 9$	J
4	L	11	K	10	$(11 + 10) \bmod 26 = 21$	V
5	O	14	E	4	$(14 + 4) \bmod 26 = 18$	S

Therefore:

Plaintext: HELLO

Keyword: KEYKE

Ciphertext: RIJVS

Step-by-step process

Step 1: Write the plaintext:

HELLO

Step 2: Repeat the keyword until its length matches the plaintext:

KEYKE

Step 3: Convert each letter into its numerical value.

Step 4: Add the corresponding plaintext and keyword values and take modulo 26.

Step 5: Convert the resulting values back into letters.

Thus:

HELLO → **RIJVS**

Effect of the repeating keyword

The repeating keyword determines which substitution is applied to each plaintext character.

For example:

- **H** is combined with **K** → **R**
- **E** is combined with **E** → **I**
- **L** is combined with **Y** → **J**
- **L** is combined with **K** → **V**
- **O** is combined with **E** → **S**

The keyword **KEY** repeats as **KEYKEY...**, so the same keyword letters are reused for later characters. This means the same plaintext letter can produce different ciphertext letters depending on the corresponding keyword letter, making the cipher stronger than a simple monoalphabetic substitution.

4. Scenario: Suspicious Digital Asset Investigation

The situation described is an example of **image steganography**, where confidential information is hidden inside an apparently normal image.

How investigators can detect hidden data

Investigators can use several forensic techniques:

1. Visual and statistical analysis

The image can first be examined for unusual patterns, distortions, or unexpected color variations. Investigators can compare the image with an original copy, when available.

2. Metadata examination

Tools such as **ExifTool** can inspect metadata such as:

- File creation/modification information
- Software used to create the file
- Unusual comments or embedded fields
- Unexpected metadata entries

3. Histogram analysis

A histogram of the image can reveal unusual distributions in RGB channels. Abnormal statistical patterns may indicate that data has been embedded.

4. Bit-plane analysis

Images can be separated into individual bit planes. Hidden information in the least significant bits (LSBs) may produce detectable patterns when the bit planes are examined.

5. Steganalysis tools

Specialized forensic tools can test whether a file contains hidden data and, in some cases, identify the likely steganography method.

Two techniques to strengthen hidden communication

Technique 1 – Encrypt the hidden information before embedding

The confidential message should first be encrypted using a strong encryption algorithm such as **AES**. Even if an investigator extracts the hidden data, the extracted information will still be unreadable without the encryption key.

Technique 2 – Use key-based randomized embedding

Instead of placing hidden bits in predictable pixel locations, the sender can use a secret key to select embedding positions pseudo-randomly. This makes statistical detection and extraction more difficult.

Overall process

Confidential message → Encryption → Randomized embedding → Image

At the receiver:

Image → Extraction using secret key → Decryption → Original message

Conclusion

The first scenario demonstrates the Vigenère cipher, where the repeating keyword controls the substitution applied to each character. The second scenario demonstrates steganography and steganalysis, where investigators analyze an image for hidden information using statistical, metadata, and bit-level techniques. Encrypting the hidden message and using key-controlled embedding can provide additional protection.